

Royal Malware Simulator - Technical Report

Executive Summary

The Royal Malware Simulator is an educational web-based application designed to demonstrate the behavioral characteristics of three primary malware types: Viruses, Worms, and Trojans. This tool provides a safe, controlled environment for cybersecurity education without any actual malicious functionality.

Project Overview

Project Name: YASHRAJSINH RATHOD ZANZMER Royal Malware Simulator

Purpose: Educational demonstration of malware propagation patterns

Technology Stack: HTML5, CSS3, Vanilla JavaScript

Target Audience: Students, educators, cybersecurity enthusiasts

System Architecture

1. Frontend Components

Visual Interface

Theme: Royal purple and gold color scheme with glassmorphism effects

Layout: Responsive grid system supporting multiple viewport sizes

Animations: CSS keyframe animations for floating crown and pulsing infected cells

Interactive Elements

Four control buttons for initiating simulations and system reset

Three independent 8x8 grids (64 cells each) representing different environments

Real-time statistics panels for each malware type

Scrollable activity log with color-coded entries

Malware Simulation Models

Virus Simulator

Behavior Pattern:

Requires host file to initiate infection

Spreads through adjacent cells (simulating file-to-file contact)

30% probability of infection per neighbor contact

Update interval: 800ms

Key Characteristics:

Dependency on host execution

Slower, controlled spread pattern

Neighbor-based propagation algorithm

Educational Value: Demonstrates why viruses need user interaction and file execution to spread

Worm Simulator

Behavior Pattern:

Self-replicating without host requirement

Dual propagation method:

Adjacent cell infection (50% probability)

Random network jump capability (20% probability)

Update interval: 600ms (faster than virus)

Key Characteristics:

Autonomous spreading mechanism

Network-aware behavior simulation

Rapid infection rate

Educational Value: Shows how worms exploit network vulnerabilities and spread independently

Trojan Simulator

Behavior Pattern:

Randomly selects 8 cells as disguised threats

Sequential activation (one per second)

No replication or spreading

Update interval: 1000ms

Key Characteristics:

Deception-based compromise

No propagation mechanism

User-triggered activation simulation

Educational Value: Illustrates social engineering and disguised threat tactics

Technical Implementation

Data Structures

```
grids = {  
    virus: [], // Array of 64 cell objects  
    worm: [], // Array of 64 cell objects  
    trojan: [] // Array of 64 cell objects  
}
```

```
Cell Object = {  
    element: DOMElement,  
    infected: boolean  
}
```

Core Algorithms

1. Neighbor Detection Algorithm

Calculates grid position (row/column)

Returns adjacent cells (up, down, left, right)

Prevents out-of-bounds access

2. Infection Propagation

Iterates through infected cells

Applies probability-based spreading

Updates visual representation in real-time

3. Statistics Engine

Counts healthy vs infected cells

Calculates infection percentage

Updates DOM elements dynamically

Features & Functionality

Visual Feedback System

Green cells: Healthy/Clean systems

Red cells: Virus/Worm infected

Gold cells: Trojan compromised

Pulsing animation: Active infection

Activity Logging

Timestamped entries

Color-coded by malware type

Scrollable history (50 entry limit)

Detailed behavior descriptions

Statistics Dashboard

Each simulation displays:

Healthy cell count

Infected cell count

Infection rate percentage

Safety & Security Considerations

Educational Safeguards

- ✓ No actual malicious code
- ✓ Isolated browser environment
- ✓ No network requests
- ✓ No data collection
- ✓ No file system access
- ✓ Client-side only execution

Ethical Implementation

Purely visual simulation

No real vulnerabilities exploited

Cannot be weaponized

Clear educational labeling

Performance Metrics

File Size: ~10KB (single HTML file)

Load Time: < 1 second

Browser Compatibility: All modern browsers

Responsive Design: Mobile and desktop

Memory Usage: Minimal (<5MB)

User Experience

Interaction Flow

User selects malware type to simulate

Grid initializes with healthy cells

Infection begins from random starting point

Real-time visual updates show spread

Statistics update continuously

Activity log records all events

Simulation completes when criteria met

User can reset and try different scenarios

Accessibility Features

High contrast color scheme

Clear visual indicators

Readable font sizes

Descriptive labels

Keyboard accessible buttons

Educational Outcomes

Learning Objectives Achieved

Understanding propagation differences between malware types

Recognizing infection patterns visually

Comparing spread velocities across threat types

Identifying key characteristics of each malware category

Practical Applications

Cybersecurity training modules

Computer science classroom demonstrations

Self-paced learning tool

Visual aid for presentations

Future Enhancement Possibilities

Additional Malware Types: Ransomware, Spyware, Rootkits

Defense Mechanisms: Antivirus simulation, Firewall barriers

Network Topology: Different grid patterns (star, mesh, ring)

Difficulty Levels: Adjustable infection rates

Save/Export: Download simulation results

Comparative Mode: Run multiple simulations side-by-side

Technical Specifications

Browser Requirements

HTML5 support

CSS3 animations

ES6 JavaScript

Dependencies

None (pure vanilla implementation)

File Structure

Single self-contained HTML file

Embedded CSS styling

Inline JavaScript logic

Conclusion

The Royal Malware Simulator successfully achieves its educational objective by providing an engaging, visually appealing, and safe environment to learn about malware behavior. The simulation accurately represents the fundamental differences between Viruses, Worms, and Trojans while maintaining a completely harmless implementation suitable for educational settings.

The royal theme adds aesthetic appeal without compromising functionality, making cybersecurity education more accessible and enjoyable for diverse audiences.

Report Generated For: YASHRAJSINH RATHOD ZANZMER

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Classification: Educational Tool

Status: Fully Functional